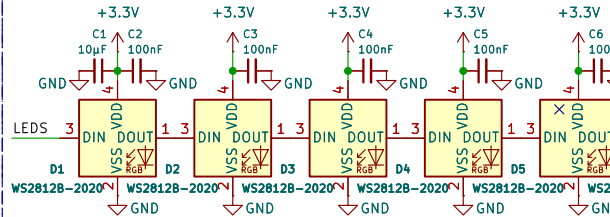


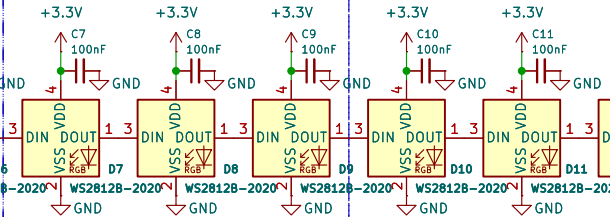
WaveShare 2.8" touch screen ESP32-S3 hat

Addressable LEDs

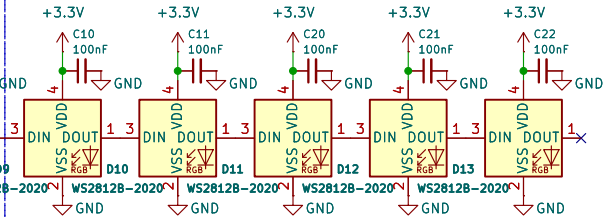
Side A



Side B

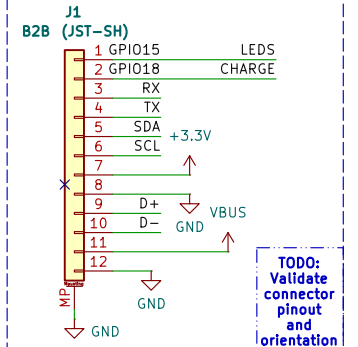


Side C

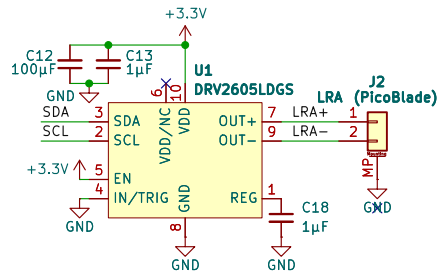


TODO:
Validate maximum LEDs, placement, and part

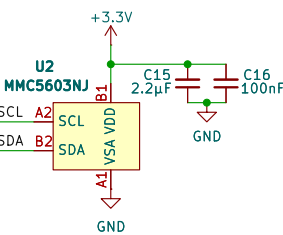
B2B Connector



Haptic Driver

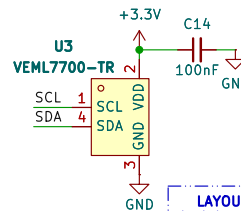


Magnetic Sensor



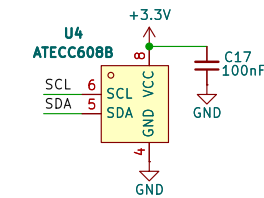
LAYOUT:
Isolate from LRA

Ambient Light Sensor

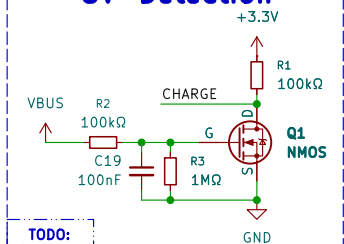


LAYOUT:
Isolate from other LEDs +
place at board edge

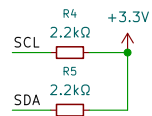
Crypto Functions



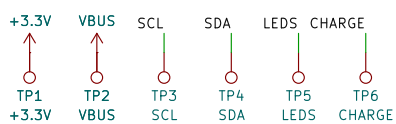
5V Detection



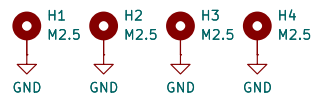
Pullups



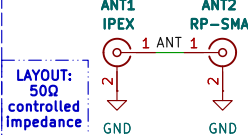
Test Points



Mounting Points



Antenna Connectors



TODO:

- Swap LEDs for C variant that supports 3.3V
- Simulate 5V detection circuit
- Calculate 50Ω controlled impedance
- Measure and source M2.5 standoffs
- Validate all LCSC Part #s and footprints
- Validate B2B connector orientation for custom cable
- Source and validate Q1 NMOS part

Beehive Engineering

Sheet: /

File: 2.8in touch display hat.kicad_sch

Title: WaveShare 2.8" touch screen ESP32-S3 hat

Size: A4

Date: 2026-08-15

KiCad E.D.A. 10.0.0

Rev: V1

Id: 1/1

